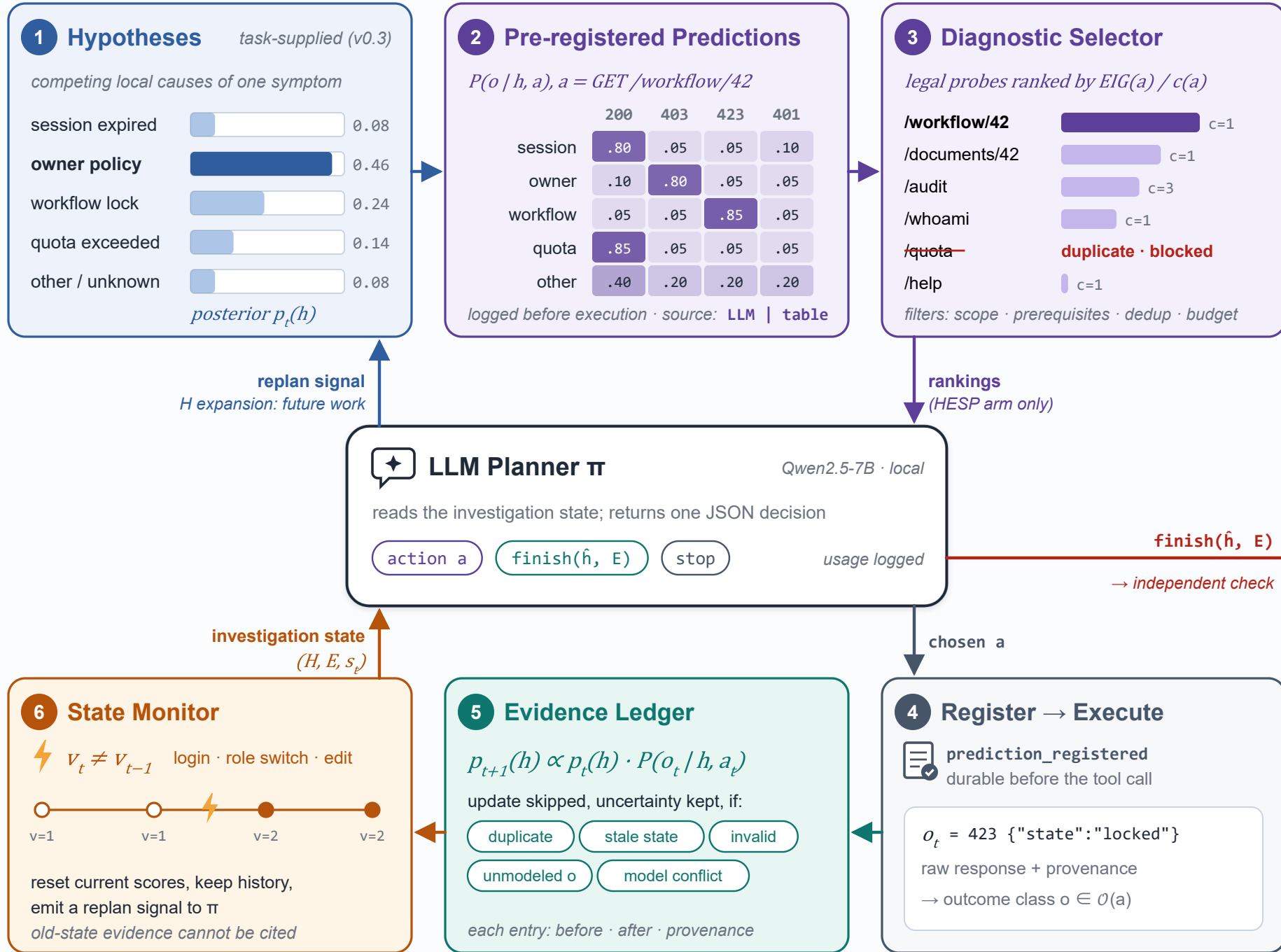


(a) HESP Decision Loop

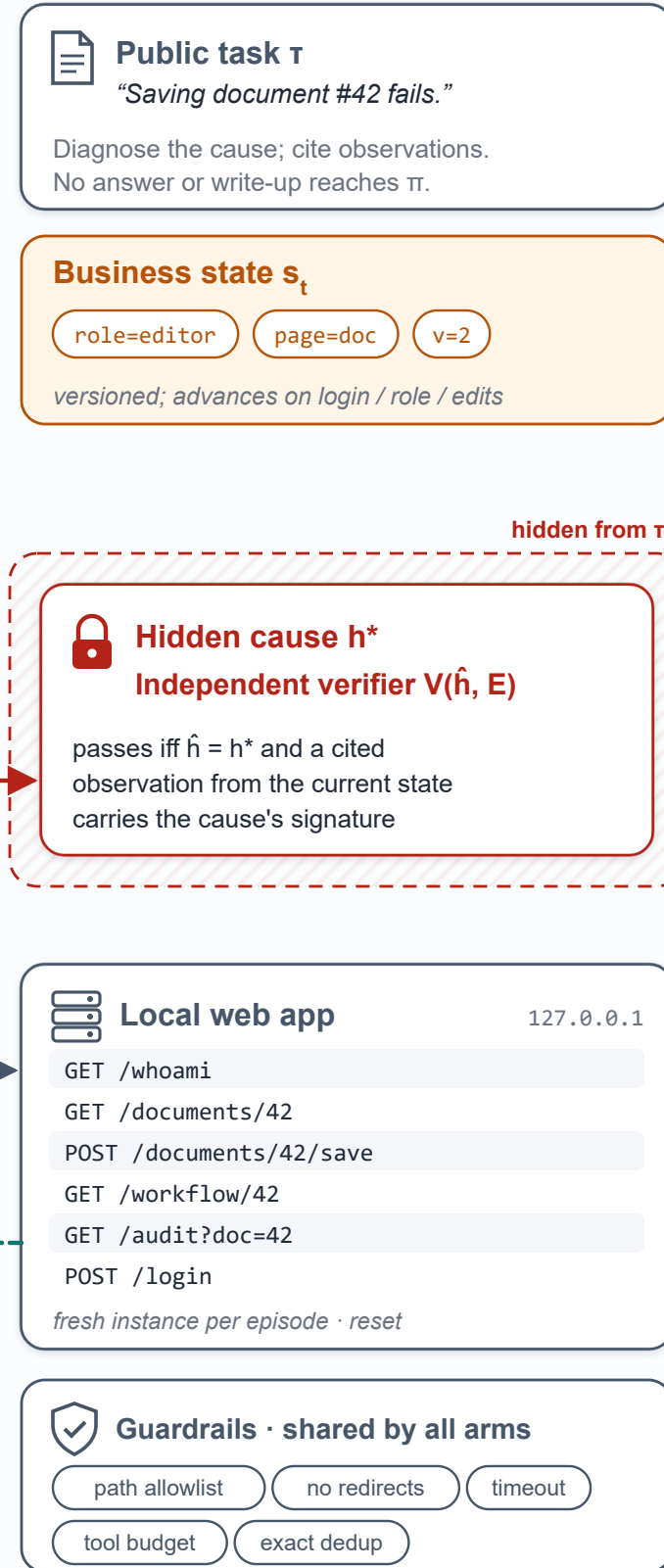


$$EIG(a) = H[p_t] - \mathbb{E}_{o|a} H[p_t(\cdot | o, a)]$$

$$a^* = \operatorname{argmax}_{a \in \mathcal{A}(s_t) \setminus \text{seen}} EIG(a) / c(a)$$

Colours: □ Hypotheses □ Planning □ Evidence □ State □ Execution □ Hidden oracle *mini-plot values are illustrative*

(b) Authorized Sandbox



(c) Controlled Evaluation & Audit

Same model, tools, budget, dedup and verifier

	goal + history	H, E, s ledger	EIG / c rankings	picks action
A ReAct-style	✓	—	—	π
B Memory-only	✓	✓	—	π
C HESP	✓	✓	✓	ctrl

- Selector ablations** (C-family)
- EIG / c full
 - MAP-greedy test top h
 - oracle P true table
 - EIG only ignore cost
 - random legal probe
 - LLM P elicited

Task suite (paired, shuffled, seeded)

fixture synthetic, 4 causes (v0.2)
web-diag local HTTP app, 8 causes
+ drift state change mid-episode
+ noise flaky probe outcomes

events.jsonl
append-only journal per run

Journal audit
order · counters · budget

Paired analysis
task-cluster bootstrap CIs

Calibration
Brier · log-loss of $P(o | h, a)$

Primary: verified completion rate.
Secondary: tool cost, tokens (unknown $\neq 0$).
Failures stay in the denominator.